

Binance Leads, but Some Wallets Anticipate: Wallet-Level Cross-Venue Informed Flow in BTC Perpetual Futures

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Abstract

This paper examines cross-venue price discovery between Binance and Hyperliquid BTC perpetual futures, asking whether informed on-chain order flow follows the dominant centralised venue or anticipates subsequent price movement on it. The study aligns Binance and Hyperliquid order-book data by venue event time over 25 May 2026 to 22 June 2026, combining lead-lag cross-correlations, Hasbrouck information-share estimates, wallet-level signed markouts, and split-sample persistence tests. Hyperliquid wallet addresses allow repeated trader-level behaviour to be observed, while Binance mid-prices provide the centralised-exchange reference series. At the venue level, Binance leads Hyperliquid in every price-discovery window, and the informed Hyperliquid cohort also follows Binance on average: signed Binance price movement before cohort trades exceeds movement after cohort trades at 2, 5 and 10 second horizons. However, the aggregate average conceals substantial wallet heterogeneity. A minority of wallets show an anticipatory signature, and split-sample tests indicate that this leadership is persistent out of sample, with the strongest half-1 lead-score quintile remaining the only quintile with positive half-2 lead across all tested horizons. The paper therefore shifts the empirical object from venue-level price discovery to wallet-level cross-venue informed flow, showing that a venue can be a follower on average while still hosting a persistent minority of wallets whose trades precede movement on the dominant venue.

Keywords: price discovery; perpetual futures; Hyperliquid; Binance; informed trading; crypto market microstructure; wallet-level data; Hasbrouck information share

1. Introduction

Price discovery in crypto derivatives markets is fragmented across centralised exchanges and on-chain order-book venues. Bitcoin perpetual futures are especially well suited to studying this fragmentation because the same economic exposure trades simultaneously across venues with very different transparency, latency, and trader-identification regimes. Centralised exchanges such as Binance offer deep order books and dense quote updates, but trader identities are not observable to outside researchers. On-chain venues such as Hyperliquid expose wallet addresses and therefore allow repeated trader-level behaviour to be measured. This creates a rare opportunity to connect the venue-level question of price discovery with the trader-level question of who appears to trade ahead of whom.

The central empirical question in this paper is whether informed trading on Hyperliquid leads or follows Binance price discovery in BTC perpetual futures. The question is deliberately two-sided. If Binance is the deeper and faster reference venue, Hyperliquid order flow may mainly react to Binance moves. If, however, some Hyperliquid wallets possess superior information, inventory positioning, or cross-venue execution capability, their trades may precede subsequent Binance price movement. The paper therefore distinguishes between aggregate venue leadership and wallet-level anticipatory behaviour.

The data combine BTC perpetual trades and book snapshots from Hyperliquid with Binance perpetual order-book snapshots over 25 May 2026 to 22 June 2026. The Hyperliquid sample contains approximately 14.0 million BTC trades across the collection span, and the analysis uses event-time alignment rather than receipt-time alignment. This distinction matters because Binance receive-lag is approximately zero at the median, while Hyperliquid receive-lag is approximately

479 milliseconds with a tail near two seconds. Aligning each venue on its own event clock avoids attributing transport delay to economic price discovery.

The venue-level evidence is clear. Across four event-time-aligned windows, Binance leads Hyperliquid. Peak cross-correlations occur at the Binance-leading grid point in every window, and Hasbrouck information-share estimates assign the majority of permanent price innovation to Binance under both Cholesky orderings. The information-share bounds are wide, so the paper avoids a falsely precise point estimate. The defensible inference is directional: Binance is the dominant price-discovery venue, while Hyperliquid contributes a real but minority share of the permanent price process.

The wallet-level evidence adds the main contribution. The paper first constructs a market-adjusted informedness measure using signed forward-return markouts on Hyperliquid and forms an informed cohort of 658 persistent wallets. At the pooled cohort level, the result still points to Binance leadership. Signed Binance mid-price movement before cohort trades is larger than movement after cohort trades at 2, 5 and 10 second horizons. In other words, the informed Hyperliquid cohort, as a whole, follows Binance.

This aggregate result, however, hides meaningful heterogeneity. Among 644 active cohort wallets, 156 wallets have a positive lead tilt and 59 wallets satisfy a stricter clean-anticipator definition. The split-sample tests then show that the leadership signal is not merely in-sample sorting. Across 588 wallets with sufficient activity in both halves, Spearman rank correlations between half-1 and half-2 lead scores are +0.477, +0.548, and +0.532 at 2, 5 and 10 second horizons. The strongest half-1 quintile remains the only quintile with positive half-2 lead across all horizons.

The paper's contribution is therefore not the claim that Hyperliquid generally leads Binance. The more precise claim is that Binance dominates aggregate price discovery, but Hyperliquid's wallet-level transparency reveals a persistent minority of anticipatory traders. This distinction matters for market microstructure research because venue-level averages can obscure trader-level information flows. It also matters for crypto market surveillance, where on-chain trader identities can be observed repeatedly but may not map cleanly to legal entities or strategies.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature on price discovery, informed trading, and crypto market microstructure. Section 3 describes the data, clock alignment, venue-level price-discovery tests, wallet-level informedness measures, and split-sample leadership tests. Section 4 reports the empirical results. Section 5 discusses interpretation, limitations, and implications. Section 6 concludes.

2. Literature review

2.1 Price discovery across fragmented markets

The first strand of literature studies how multiple venues trading the same or closely related asset contribute to a common efficient price. In fragmented equity and futures markets, no single transaction series necessarily reveals the full price-discovery process. Hasbrouck (1995) provides a widely used information-share framework that attributes permanent price innovations across cointegrated markets. Gonzalo and Granger (1995) offer a related permanent-transitory decomposition. Both approaches motivate the empirical separation between a common efficient price and venue-specific temporary deviations.

This paper follows that tradition by treating Binance and Hyperliquid BTC perpetual mids as prices for the same underlying economic exposure. The venue-level analysis asks which venue contributes more to permanent price innovation. Lead-lag cross-correlation provides a simple directional diagnostic, while Hasbrouck information shares quantify the contribution of each venue to the common factor. Because the two venues have different snapshot frequencies, the paper treats the measured lag as grid-bounded rather than literal. The preferred grid is chosen near Hyperliquid's effective book frequency to avoid stale-filling artefacts that would mechanically favour the denser Binance series.

2.2 Informed trading and trader heterogeneity

The second strand concerns informed trading and adverse selection. Classic models such as Kyle (1985) and Glosten and Milgrom (1985) show how informed traders can move prices through order flow, while market makers adjust quotes in response to adverse-selection risk. Empirical market microstructure has developed several ways to detect informed

trading, including signed return markouts, order-flow toxicity measures such as those of Easley, Lopez de Prado and O'Hara (2012), and the persistence of trader or account-level performance (Easley, Kiefer, O'Hara and Paperman, 1996). Broader surveys of these microfoundations and empirical methods are provided by O'Hara (1995), Madhavan (2000) and Biais, Glosten and Spatt (2005).

Most datasets used in traditional markets do not allow researchers to observe persistent trader identities. As a result, empirical work often measures informedness at the venue, security, account class, or aggregate order-flow level. On-chain order books change the empirical setting. Wallet addresses are not perfect identities, but they do allow the same address to be followed across repeated trades. This makes it possible to distinguish between aggregate informed flow and heterogeneous wallet-level behaviour. The present paper exploits that feature by constructing wallet-level markouts, selecting a persistent informed cohort, and then testing whether leadership survives out of sample.

2.3 Crypto perpetual futures and on-chain transparency

Crypto markets provide a natural environment for cross-venue price-discovery tests because assets trade continuously across many venues with heterogeneous market designs. Prior work on cryptocurrency markets documents segmentation and arbitrage frictions (Makarov and Schoar, 2020), manipulation risk (Gandal, Hamrick, Moore and Oberman, 2018; Griffin and Shams, 2020), and the importance of venue structure. Brauneis and Mestel (2018) study price discovery and efficiency across cryptocurrencies, while Alexander, Choi, Park and Sohn (2020) show that Bitcoin derivatives can lead spot-market prices. Perpetual futures intensify these issues because they are highly liquid, frequently used for leverage, and directly linked to spot and derivative price discovery.

The pairing of Binance and Hyperliquid is especially informative. Binance represents the deep centralised venue with dense order-book data. Hyperliquid is an on-chain order-book venue where trades carry buyer and seller wallet addresses. The empirical contribution of this paper is to combine the two: Binance provides the benchmark venue for price discovery, while Hyperliquid provides trader-level observability. The result is a cross-venue design that can ask not only whether one venue leads another, but whether particular wallets on the follower venue display persistent anticipatory behaviour.

3. Data and method

3.1 Data and clock alignment

The analysis uses BTC perpetual data from two venues: Binance, a centralised exchange, and Hyperliquid, an on-chain order book. For venue v in $\{B, H\}$, where B denotes Binance and H denotes Hyperliquid, the mid-price at time t is defined as the average of the best ask and best bid:

$$m_{t^v} = 0.5(a_{t^v} + b_{t^v}). \quad (1)$$

Each Hyperliquid trade records an event timestamp, executed price, size, aggressor direction, and the buyer and seller wallet addresses. Binance book snapshots arrive at roughly one-second spacing, while Hyperliquid book snapshots arrive at roughly four-second spacing. The sample window runs from 25 May 2026 to 22 June 2026. The full collection span contains 14,001,692 Hyperliquid BTC trades, or approximately 0.48 million trades per day.

The venues are aligned on their own event timestamps rather than receipt timestamps. Let l_i^v denote receive-lag, the difference between receipt and event timestamps. Binance's median receive-lag is approximately zero milliseconds, while Hyperliquid's median receive-lag is approximately 479 milliseconds, with a tail near two seconds. Event-time alignment avoids importing Hyperliquid's transport delay into the cross-venue comparison. When a venue mid is required at an arbitrary time t , the most recent prior snapshot is used:

$$m^v(t) = m^v_{\{t_k\}}, \quad \text{for } t \text{ in } [t_k, t_{k+1}). \quad (2)$$

This right-continuous step-function treatment matches the economic availability of the most recent book update. It also avoids interpolation across discrete order-book states.

Table 1. Data and sample

Item	Detail
DEX venue	Hyperliquid (on-chain perpetual order book; trades carry buyer/seller wallet addresses)

Item	Detail
CEX venue	Binance (perpetual order book)
Instrument	BTC perpetual
Sample window	25 May 2026 - 22 Jun 2026 (29 days)
Hyperliquid BTC trades	14,001,692 over full collection span; approx. 0.48 million/day
Hyperliquid book snapshots	Approx. 4 s spacing
Binance book snapshots	Approx. 1 s spacing
Clock alignment	Each venue aligned on its own event_ts_ms; Binance median receive-lag approx. 0 ms; Hyperliquid median approx. 479 ms with tail near 2 s

Note: The event-clock design is central to the cross-venue inference.

3.2 Venue-level price discovery

The venue-level analysis treats the two log mid-price series as cointegrated prices for the same economic exposure. Price discovery is measured in two complementary ways. First, the two mid-price series are sampled on a common grid $\tau_j = \tau_0 + j \Delta$, and grid log-returns are defined as:

$$r_j^v = \ln m^v(\tau_j) - \ln m^v(\tau_{j-1}). \quad (3)$$

The lead-lag cross-correlation at lag ℓ is:

$$\rho(\ell) = \text{Corr}(r_{j+\ell}^B, r_j^H). \quad (4)$$

Under this convention a negative reported lag indicates that Binance moved one or more grid intervals before Hyperliquid, so a peak at minus five seconds means Binance-leading alignment is strongest at the five-second grid. The reported lag should be interpreted as directional and grid-bounded rather than a literal five-second economic lead, because the five-second grid cannot resolve a sub-five-second lead.

Second, the paper estimates Hasbrouck information shares. Let $p_j = (\ln m_j^B, \ln m_j^H)'$ and $\beta = (1, -1)'$. The vector error-correction model is:

$$\begin{aligned} \Delta p_j &= \alpha \beta' p_{j-1} + \sum_{i=1}^L \Gamma_i \Delta p_{j-i} + \epsilon_j, \\ \text{Cov}(\epsilon_j) &= \Omega. \end{aligned} \quad (5)$$

Let ψ be the row vector orthogonal to the error-correction loadings α , so that $\psi \alpha = 0$. If F is the Cholesky factor of Ω , with $FF' = \Omega$, the Hasbrouck information share for venue v is:

$$IS_v = ([\psi F]_v)^2 / (\psi \Omega \psi'). \quad (6)$$

Because the residual covariance matrix is generally non-diagonal, the Cholesky factorisation depends on ordering. The paper therefore reports lower and upper bounds from the two orderings. The preferred estimation uses a five-second grid, chosen near the effective Hyperliquid book frequency. A 250 millisecond grid is excluded from inference because it stale-fills the slower Hyperliquid book and produces an implausible Binance share of approximately 99.7 percent.

3.3 Wallet-level informedness and cohort construction

Informedness is measured with signed forward-return markouts on Hyperliquid. For trade i at horizon h , with aggressor direction $d_i = +1$ for buyer-initiated trades and $d_i = -1$ for seller-initiated trades, the signed markout is:

$$M_i(h) = d_i [(\ln m(t_{i+h}) - \ln m(t_i)) / \ln m(t_i)] \times 10,000. \quad (7)$$

The factor of 10,000 converts the decimal return to basis points, so markouts are reported in basis points throughout. A positive markout means that the Hyperliquid mid subsequently moved in the direction of the trade. To reduce the risk of classifying market drift as trader skill, each markout is adjusted by removing the direction-signed average market move over the same day and horizon:

$$\tilde{M}_i(h) = M_i(h) - d_i \bar{g}_{\text{day}}(i)(h). \quad (8)$$

Here $\bar{g}_{\text{day}}(h)$ denotes the unconditional same-day horizon- h mid-price return, expressed in basis points. The adjustment removes common same-day directional drift before ranking wallets by informedness, so that a wallet that merely trades with same-day drift has an adjusted markout near zero.

$$\bar{M}_{\tilde{w}}(h) = (1/n_w) \sum_{i \in w} \tilde{M}_i(h). \quad (9)$$

Per-wallet informedness is the mean adjusted markout in (9), computed using streaming running sums. The sample is split into two halves and the rank correlation of wallet markouts across halves is used to test persistence. The informed

cohort C consists of persistent wallets with sufficient trades in each half, ranked by full-sample adjusted markout, taking the top quintile. This yields 658 wallets from 3,294 persistent wallets, with cohort adjusted markout from +1.15 to +22.92 basis points.

3.4 Cross-venue lead test

The main cross-venue test asks whether cohort trades lead or follow Binance. A cohort trade is one in which exactly one of the two Hyperliquid wallets belongs to C. The trade direction d_i is +1 if the cohort wallet is the buyer and -1 if the cohort wallet is the seller. For each cohort trade, the signed Binance mid-price move before and after the trade is:

$$\text{pre}_i(h) = d_i [(m^B(t_i) - m^B(t_i-h)) / m^B(t_i-h)] \times 10,000. \quad (10)$$

$$\text{post}_i(h) = d_i [(m^B(t_i+h) - m^B(t_i)) / m^B(t_i)] \times 10,000. \quad (11)$$

The cohort-level statistics are the means of $\text{pre}_i(h)$ and $\text{post}_i(h)$ over all cohort trades. If the mean post-trade move exceeds the mean pre-trade move and is positive, the cohort leads Binance. If the mean pre-trade move exceeds the mean post-trade move, the cohort follows Binance. As a placebo, the direction labels are randomly permuted and the mean post-trade statistic is recomputed; under no signed relationship, this statistic should be close to zero.

3.5 Per-wallet leadership and split-sample persistence

The cohort statistic is decomposed to the wallet level. For wallet w , $\text{pre}_w(h)$ and $\text{post}_w(h)$ are the means of equations (10) and (11) over the wallet's cohort trades. The wallet lead score is:

$$L_w(h) = \text{post}_w(h) - \text{pre}_w(h). \quad (12)$$

A wallet has a lead tilt if $\text{post}_w > \text{pre}_w$ and $\text{post}_w > 0$. A wallet is classified as a clean anticipator if $\text{pre}_w < 0.3$ bps and $\text{post}_w > 0.5$ bps. The stricter definition is conservative because a high lead score can be produced mechanically by a strongly negative pre-trade term; the clean-anticipator rule requires little Binance movement before the Hyperliquid trade and clear movement after it.

Because ranking several hundred wallets and reading the top of the distribution is vulnerable to overfitting, leadership is validated out of sample. The window is split into halves. For wallets with at least 100 cohort trades in each half, lead scores are computed separately in half 1 and half 2. The paper reports Spearman rank correlations, sorts wallets into half-1 lead-score quintiles, and measures mean half-2 lead in each quintile. The anticipator definition is also applied in half 1 and the same wallets are tracked into half 2. The procedure is repeated at 2, 5 and 10 second horizons.

4. Results

4.1 Venue-level price discovery

Table 2 reports the venue-level price-discovery diagnostics. Binance leads Hyperliquid in every event-time-aligned window. The peak cross-correlation occurs at the Binance-leading grid point in all four windows, with peak values ranging from +0.497 to +0.732. Contemporaneous correlations are positive in all windows, ranging from +0.199 to +0.417, indicating that the two venues share a common price process even though leadership is asymmetric.

The Hasbrouck results point in the same direction. Hyperliquid's information-share midpoint is economically meaningful but minority-sized, while Binance holds the majority share under both Cholesky orderings in every window. The bounds are wide, particularly in the one-hour windows, because the five-second grid gives only about 720 observations per hour and the residual correlation leaves the two orderings far apart. The paper therefore emphasises the robust directional finding - Binance leads and holds the majority information share - rather than a falsely precise Hyperliquid percentage.

Two implementation choices are important for interpretation. First, the peak at -5 seconds should not be read as a literal five-second lead. The grid cannot resolve sub-five-second leadership, so the lag identifies direction rather than exact duration. Second, the excluded 250 millisecond grid is not a failed robustness result but a design diagnostic: it over-represents Binance because Hyperliquid's slower book is stale-filled too aggressively. The preferred five-second grid better matches the effective Hyperliquid book frequency.

Table 2. Cross-venue price discovery: lead-lag and information share

Window	Span	Peak cross-corr (lag)	Contemp. corr	Hyperliquid IS [lower, upper]	Binance IS [lower, upper]
1	20 Jun, 1 h	+0.518 (-5 s)	+0.378	[2.7%, 40.9%]	[59.1%, 97.3%]
2	12 Jun, 1 h	+0.624 (-5 s)	+0.417	[0.9%, 44.5%]	[55.5%, 99.1%]
3	16 Jun, 1 h	+0.732 (-5 s)	+0.199	[0.0%, 14.6%]	[85.4%, 100.0%]
4	15 Jun, 6 h	+0.497 (-5 s)	+0.278	[3.3%, 23.0%]	[77.0%, 96.7%]

Note: Four event-time-aligned windows, five-second grid. Hasbrouck shares are Cholesky-ordering bounds. The lag indicates Binance-leading direction at the grid resolution, not an exact measured lead length.

4.2 Wallet-level informedness

The next step is to identify informed Hyperliquid wallets before testing cross-venue leadership. Table 3 summarises the wallet-level informedness diagnostics. The full-window analysis covers approximately 13.2 million trades. The one-day pilot markout spread between the 90th and 10th percentiles widens from 1.85 bps at 10 seconds to 6.05 bps at 300 seconds, indicating that trader-level outcomes are meaningfully dispersed and grow with horizon.

Market adjustment matters. The top informed wallets after adjustment have sell shares between 0.35 and 0.70, suggesting that pure directional names are removed by the adjustment rather than mistaken for skill. Market-adjusted makers are slightly negative, consistent with mild adverse selection but also with hedged liquidity provision. Informedness is persistent but modest: the half-1 versus half-2 Spearman correlation is +0.141, with p approximately zero. Out-of-sample persistence is examined in detail at the leadership level in Section 4.5 and Table 6.

Table 3. Wallet-level informedness and its persistence

Quantity	Value
Trades analysed (full window)	13.2 million
Signed forward-return markout, p90 - p10 spread (1-day pilot)	1.85 → 2.20 → 3.14 → 6.05 bps at 10 / 30 / 60 / 300 s
Top informed wallets after market adjustment	Sell-share range 0.35 - 0.70
Makers (market-adjusted)	Slightly negative
Persistence of informedness	Spearman rho = +0.141, p approx. 0

Note: Informedness is persistent but modest in magnitude; the half-1 versus half-2 rank correlation is positive and significant. The complete out-of-sample leadership-persistence ladder is reported in Table 6.

4.3 Aggregate informed cohort follows Binance

The informed cohort consists of 658 wallets, the top quintile of 3,294 persistent wallets by market-adjusted markout. The cohort contributes 2,337,051 cohort trades to the cross-venue lead test. Table 4 and Figure 1 show the pooled result. At every horizon, signed Binance movement before the cohort trade is larger than signed Binance movement after the cohort trade. At the two-second horizon, Binance moves +1.731 bps before the cohort trade and +0.604 bps after it. At five seconds, the corresponding values are +2.043 bps and +0.654 bps. At ten seconds, they are +2.371 bps and +0.720 bps.

The interpretation is direct. The informed Hyperliquid cohort follows Binance on average. The result does not imply that the cohort is uninformed in the Hyperliquid market. Rather, it implies that its cross-venue timing, when pooled, is reactive to Binance price discovery. The shuffled-direction placebo reinforces the signed interpretation: the five-second placebo post statistic is +0.006 bps, essentially zero.

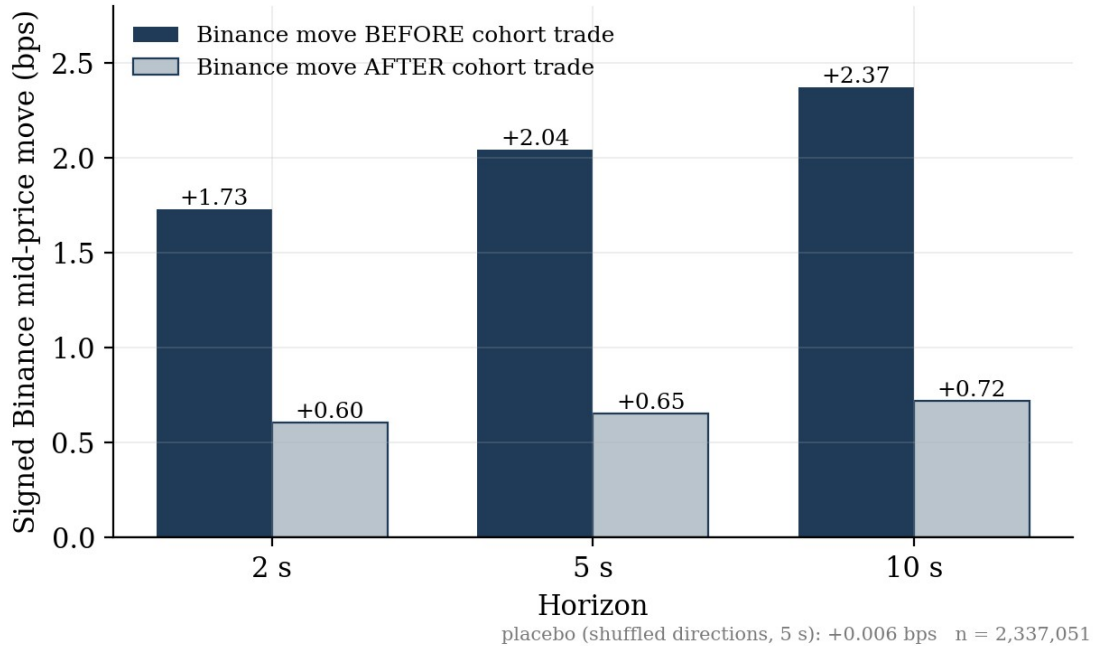
Table 4. Cross-venue lead test: pooled informed cohort

Horizon	Binance move BEFORE (pre)	Binance move AFTER (post)	Reading
2 s	+1.731 bps	+0.604 bps	Follows
5 s	+2.043 bps	+0.654 bps	Follows
10 s	+2.371 bps	+0.720 bps	Follows
Placebo, shuffled directions, 5 s	-	+0.006 bps	Approx. zero

Note: Cohort = 658 wallets; n = 2,337,051 cohort trades. Pre dominates post at every horizon.

Figure 1. Aggregate informed cohort follows Binance.

Aggregate informed cohort follows Binance



4.4 Per-wallet leadership heterogeneity

The pooled average masks substantial heterogeneity. Of the 658 cohort wallets, 644 have at least 200 cohort trades and enter the per-wallet distribution. Table 5 decomposes the five-second lead test across these 644 wallets. The median lead score is -0.69 bps, consistent with the aggregate following result, but the distribution is wide: from -17.54 bps to +12.43 bps. More importantly, 156 of 644 wallets, or 24 percent, show a lead tilt, defined as $\text{post} > \text{pre}$ and $\text{post} > 0$. A stricter set of 59 wallets, or 9 percent, satisfies the clean-anticipator definition of $\text{pre} < 0.3$ bps and $\text{post} > 0.5$ bps.

The distinction between lead score and clean anticipation is important. A high lead score can be produced by a strongly negative pre-trade term, even if the post-trade movement is weak. The clean-anticipator rule is designed to avoid this mechanical interpretation. It requires little Binance movement before the Hyperliquid trade and clearly positive Binance movement after the trade. The named wallets in Table 5 should therefore be read as descriptive wallet identifiers, not as claims about legal entities or intentions.

Table 5A. Per-wallet leadership distribution

Statistic	Value
Wallets with a lead tilt	156 / 644 (24%)
Clean anticipators	59 / 644 (9%)
Lead-score distribution	min / median / max = -17.54 / -0.69 / +12.43 bps

Note: Five-second horizon. Lead score equals post minus pre.

Table 5B. Top and bottom wallets by five-second lead score

Wallet	n	pre (bps)	post (bps)	lead (bps)	Reading
0x77eeda19	2,556	+0.95	+13.38	+12.43	Top lead score
0xa516541e	1,904	-8.23	+0.29	+8.52	High lead score from negative pre
0x621c5551	1,600	-9.34	-0.87	+8.46	High lead score from negative pre
0xc04f3944	268	-4.33	+2.40	+6.73	Positive post
0x50a4d39b	1,323	-3.04	+1.52	+4.55	Positive post
0x32008fcb	1,610	-0.38	+3.82	+4.20	Clean-style anticipator
0x732f7178	911	-1.04	+3.16	+4.20	Positive post
0x9137bd93	3,672	+0.08	+4.20	+4.12	Clean-style anticipator
0x0ab8a677	2,961	-2.89	+1.34	+4.23	Positive post
0x3faf35a2	463	+16.77	-0.77	-17.54	Strong follower
0x66701345	354	+18.33	+1.36	-16.97	Strong follower
0x42be8a7a	581	+16.14	-0.00	-16.14	Strong follower

Note: Wallet addresses are persistent identifiers in the data, not legal identity claims.

4.5 Leadership persists out of sample

The per-wallet distribution motivates a stronger validation test. If leadership were merely the result of sorting hundreds of wallets in sample, it should weaken or disappear out of sample. Table 6 and Figures 2 and 3 show that this is not the case. Among 588 wallets with at least 100 cohort trades in both halves, half-1 and half-2 lead scores are strongly rank-correlated. Spearman correlations are +0.477 at the two-second horizon, +0.548 at the five-second horizon, and +0.532 at the ten-second horizon.

The quintile ladder is also economically intuitive. Wallets in the weakest half-1 lead quintile remain strongly negative in half 2. The middle quintiles improve but remain negative. Only Q5, the strongest half-1 lead-score quintile, is positive out of sample at all horizons, with half-2 leads of +0.45, +0.78, and +0.73 bps at 2, 5 and 10 seconds. This is the paper's central wallet-level finding: leadership is a stable trait for the top-ranked group, even though the effect is modest in basis-point terms.

The anticipator-versus-benchmark comparison reinforces the conclusion. Half-1 anticipators generate mean half-2 leads of +0.70, +0.75, and +0.79 bps across the three horizons. The non-anticipator benchmark is negative at -1.04, -1.56, and -2.21 bps. Roughly half of the first-half anticipators remain individually leading in half 2. This persistence rate should not be overstated as single-wallet prediction, but it is strong evidence that the anticipator group is not a random in-sample artefact.

Table 6. Lead persistence: split-sample tests across horizons

Quantity	2 s	5 s	10 s
Spearman rho (half-1 vs half-2 lead)	+0.477	+0.548	+0.532
Q1 half-2 lead (bps)	-4.29	-5.78	-6.87
Q2 half-2 lead (bps)	-0.36	-1.03	-1.70
Q3 half-2 lead (bps)	-0.19	-0.39	-0.90
Q4 half-2 lead (bps)	-0.06	-0.26	-0.64
Q5 half-2 lead (bps)	+0.45	+0.78	+0.73
Half-1 anticipators (n)	49	57	65
Their half-2 mean lead (bps)	+0.70	+0.75	+0.79
Still leading in half 2	25/49 (51%)	32/57 (56%)	33/65 (51%)
Non-anticipator benchmark (bps)	-1.04	-1.56	-2.21

Note: Wallets ranked in half 1 and evaluated in half 2. Sample includes 588 wallets with at least 100 cohort trades in each half.

Figure 2. Out-of-sample persistence: half-1 leaders stay ahead in half 2.

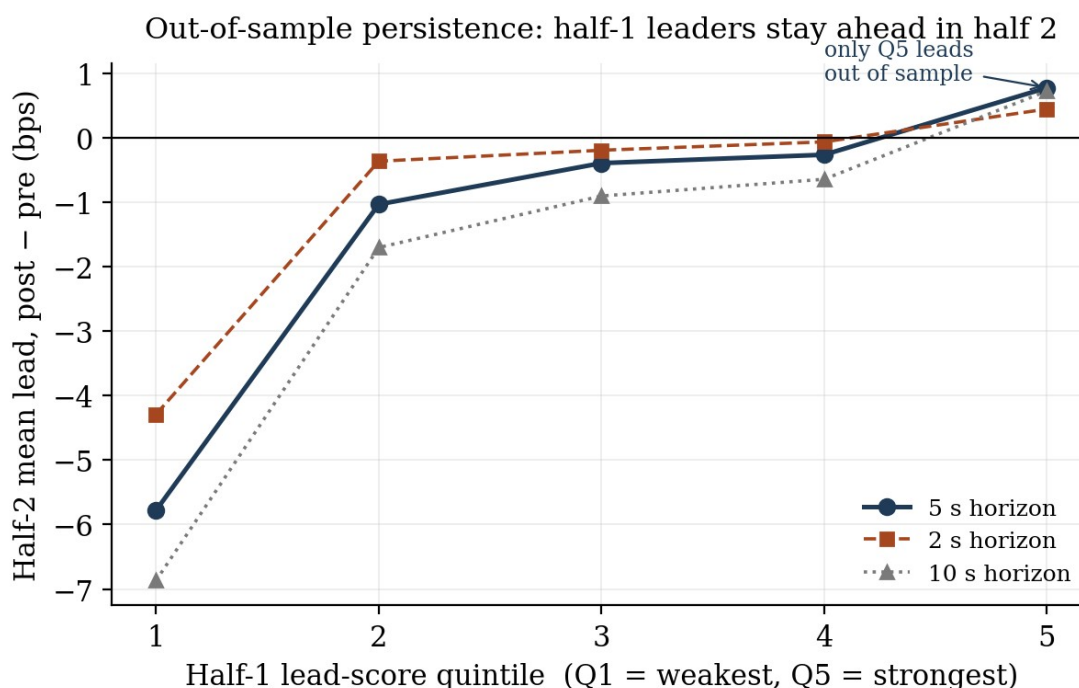
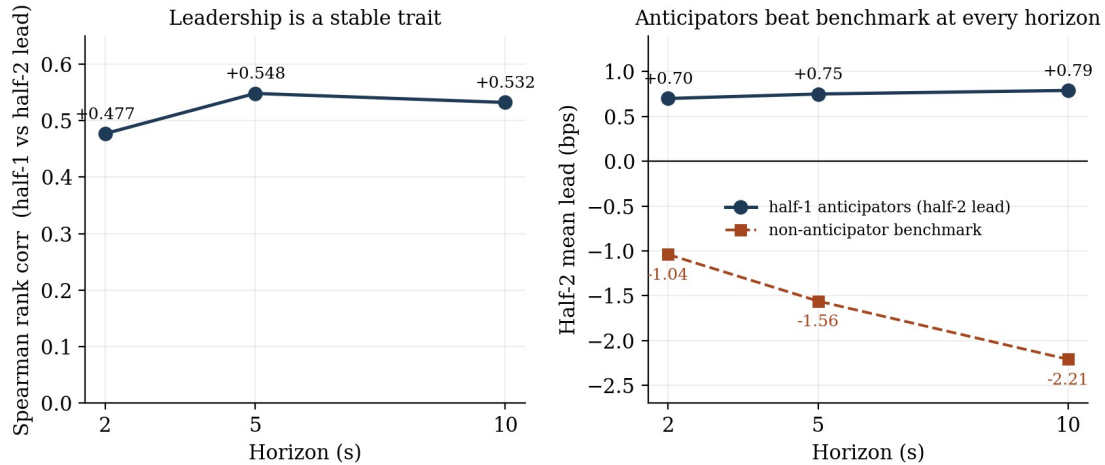


Figure 3. Robustness across 2, 5, and 10 second horizons.



5. Discussion

The results support a layered interpretation of cross-venue price discovery. At the aggregate venue level, Binance leads. At the pooled informed-cohort level, Hyperliquid follows Binance. At the wallet level, however, a persistent minority of Hyperliquid wallets anticipates subsequent Binance movement. These statements are not contradictory. They show that venue-level averages can hide trader-level heterogeneity.

The first implication is methodological. Standard price-discovery analysis asks which venue contributes more to the common efficient price. That question remains important, and the answer here is Binance. But on-chain trader identifiers allow a second question: within the venue that follows on average, are there wallets whose flow carries forward-looking information? The split-sample results indicate that there are. The empirical object therefore shifts from venues alone to the joint distribution of venue leadership and wallet-level leadership.

The second implication concerns market design and surveillance. Persistent anticipatory wallets may reflect superior information processing, faster cross-venue execution, inventory-driven trading, private order-flow information, or statistical arbitrage. The present data identify the timing pattern but do not identify the mechanism. This distinction is important. The paper does not claim manipulation, privileged information, or intent. It documents persistent predictive ordering between a subset of Hyperliquid wallets and subsequent Binance mid-price moves.

The third implication is economic scale. The anticipator effect is robust but small. The top out-of-sample quintile leads by less than one basis point on average, and the first-half anticipator group has half-2 mean leads of approximately +0.70 to +0.79 bps. These are meaningful at high frequency and high turnover, but they are not large enough to support strong claims about individual-wallet prediction. The strongest defensible claim is group-level persistence.

Several limitations should be acknowledged. First, the sample covers one instrument and one 29-day window. Results may differ for other assets, volatility regimes, or venue pairs. Second, the measured lead length is constrained by book-snapshot frequency. The five-second grid identifies Binance-leading direction but cannot measure a sub-five-second lead precisely. Third, wallet addresses are not legal identities. One trader may control multiple wallets, and one wallet may represent more than one strategy. Fourth, the clean-anticipator thresholds are fixed cut-offs designed for conservative classification, not structural estimates of information arrival. Finally, the market-adjustment and Hasbrouck specifications follow the definitions in Section 3; the information-share bounds in particular are wide and are reported as bounds rather than point estimates.

A further caveat concerns statistical inference. Because the cross-venue lead tests use high-frequency observations, trade-level counts should not be interpreted as independent observations. The relevant uncertainty is clustered by time and by wallet. The main pre-post differences, per-wallet lead scores, and split-sample quintile means are therefore best accompanied by block-bootstrap or wallet-clustered confidence intervals, and the magnitudes reported here should be read as point estimates whose sampling uncertainty is governed by the number of independent time blocks and wallets

rather than the raw trade count. This qualification is especially relevant because the out-of-sample anticipator magnitudes are below one basis point.

Despite these limitations, the evidence is internally consistent. The aggregate result says Binance dominates. The per-wallet distribution says the follower venue contains heterogeneous traders. The split-sample tests say that the strongest leadership signals persist out of sample. The combined conclusion is therefore more nuanced than a simple venue ranking: Binance leads the market, but Hyperliquid reveals who sometimes trades ahead of Binance.

6. Conclusion

This paper studies cross-venue informed flow between Binance and Hyperliquid BTC perpetual futures. Using event-time-aligned order-book data and wallet-level Hyperliquid trades, it asks whether informed on-chain flow follows the dominant centralised venue or anticipates subsequent movement on it.

The answer is both simple and subtle. Binance leads at the venue level, and the pooled informed Hyperliquid cohort follows Binance on average. However, the wallet-level data reveal a persistent minority of anticipatory wallets. Split-sample tests show strong rank persistence in lead scores, a monotone out-of-sample quintile ladder, and positive half-2 performance for first-half anticipators relative to negative non-anticipator benchmarks.

The paper's contribution is to show that venue-level price discovery and trader-level anticipation need not coincide. A venue can be a follower on average while still hosting traders whose flow precedes movement on the leading venue. This is precisely the empirical advantage of on-chain order-book data: it makes repeated trader-level behaviour observable. Future work can extend the design across assets, longer samples, and additional venue pairs, and can test whether anticipatory wallets share observable execution patterns, inventory dynamics, or cross-venue timing signatures.

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